CHL2002 · Chemical Engineering Thermodynamics · Computer Project, Fall 2026

Real Gas Calculator

PVT behaviour, residual enthalpy and entropy, and fugacity of pure gases from the generic cubic equation of state (vdW, RK, SRK, PR)

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Repository: https://github.com/sohaali1612/Real-Gas-Calculator-Project--CHL2002-Fall-2026

Note: please **run the code on a Google Colab notebook (preferred).** The program (`pvt_interactive_simulator.py`) is written for a notebook: it uses `ipywidgets` controls and the `%matplotlib inline` magic, so it will not run as a plain `python` script. In Colab, open a new notebook, paste the whole file into one code cell and run it; the fluid drop-down and the T, P, $P_{\max}$ sliders appear under the cell. NumPy, pandas, Matplotlib and `ipywidgets` are available in Colab by default.

Abstract. A notebook calculator predicts Z, molar volume, residual and total H and S, and the fugacity coefficient and fugacity of 36 pure gases from four cubic equations of state in one generic form (SVNA, Table 3.1). It reproduces the textbook benchmark of Table 13.7 (n-butane, 500 K, 50 bar) to the fourth digit, its closed-form residual properties agree with numerical integration of the defining integrals to about 10^{-5} relative or better, and PR vapour pressures at the normal boiling point are within about 1% for non-polar fluids.

1. Objective and scope

The project covers the three stages of the computer project: (i) *PVT prediction* (Z and V for the vapour-like and liquid-like roots), (ii) *enthalpy and entropy* from residual properties plus an ideal-gas path, and (iii) *pure-component fugacity*. The equations follow Smith, Van Ness, Abbott and Swihart (SVNA, 8th ed.) [1]: Sec. 3.6 and Table 3.1 for the generic cubic, Sec. 13.6 (Eqs. 13.72–13.76) for the closed-form residual properties, and Sec. 13.7 for phase equilibrium. Property constants come from Tables B.1 and C.1 of the same book.

2. Theory

2.1 Generic cubic equation of state

All four equations are special cases of SVNA Eq. (3.41):

$$P = \frac{RT}{V-b} - \frac{a(T)}{(V+\epsilon b)(V+\sigma b)}, \qquad a(T) = \Psi\,\frac{\alpha(T_r)\,R^2T_c^2}{P_c}, \qquad b = \Omega\,\frac{RT_c}{P_c}, \tag{1}$$

where $T_r = T/T_c$, $P_r = P/P_c$, and the constants $\sigma, \epsilon, \Omega, \Psi$ and the function $\alpha(T_r)$ distinguish the equations (Table 1). With $\beta = \Omega P_r/T_r$ and $q = \Psi\alpha(T_r)/(\Omega T_r)$ (SVNA Eqs. 3.50, 3.51), the equation becomes $Z = 1 + \beta - q\beta\,(Z-\beta)/[(Z+\epsilon\beta)(Z+\sigma\beta)]$ (Eq. 3.48), i.e. the cubic polynomial in Z

$$Z^3 + \big[(\epsilon+\sigma-1)\beta - 1\big]Z^2 + \big[q\beta + \epsilon\sigma\beta^2 - (\epsilon+\sigma)\beta(\beta+1)\big]Z - \big[\epsilon\sigma\beta^2(\beta+1) + q\beta^2\big] = 0. \tag{2}$$

For PR this reduces to the familiar $Z^3 - (1-B)Z^2 + (A - 3B^2 - 2B)Z - (AB - B^2 - B^3) = 0$ with $A = q\beta = aP/(RT)^2$ and $B = \beta = bP/RT$. In the (u, w) notation of the lecture notes, $u = \sigma + \epsilon$ and $w = \sigma\epsilon$, giving $(0, 0)$ for vdW, $(1, 0)$ for RK and SRK and $(2, -1)$ for PR.

Table 1: Parameter assignments (SVNA Table 3.1), as coded in the `MODELS` dictionary.

EOS	$\alpha(T_r)$	σ	ϵ	Ω	Ψ	Z_c
vdW	1	0	0	$1/8$	$27/64$	$3/8$
RK	$T_r^{-1/2}$	1	0	0.08664	0.42748	$1/3$
SRK	$\big[1+\kappa(1-\sqrt{T_r})\big]^2$, $\kappa = 0.480 + 1.574\omega - 0.176\omega^2$	1	0	0.08664	0.42748	$1/3$
PR	$\big[1+\kappa(1-\sqrt{T_r})\big]^2$, $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	0.30740

2.2 Root selection and phase

Above T_c the cubic has one real root. Below T_c it can have three: the largest is vapour-like, the smallest liquid-like, and the middle root has no physical meaning (SVNA Sec. 13.7). Only roots with $Z > \beta$ (i.e. $V > b$) are kept. The stable phase at fixed T and P is the root with the lowest G^R/RT, equivalently the lowest fugacity coefficient. The label is *Supercritical* if $T \geq T_c$, else *Vapor* if $V > Z_c RT_c/P_c$, else *Liquid*.

2.3 Residual properties and fugacity

With $I = \dfrac{1}{\sigma-\epsilon}\ln\dfrac{Z+\sigma\beta}{Z+\epsilon\beta}$ (for vdW, $I=\beta/Z$; SVNA Eq. 13.72), the residual properties are

$$\ln\varphi = \frac{G^R}{RT} = Z-1-\ln(Z-\beta)-qI, \qquad \frac{H^R}{RT} = Z-1+\left[\frac{d\ln\alpha}{d\ln T_r}-1\right]qI, \qquad \frac{S^R}{R} = \ln(Z-\beta)+\frac{d\ln\alpha}{d\ln T_r}qI, \quad (3)$$

(SVNA Eqs. 13.74–13.76), and the pure-component fugacity is $f=\varphi P$. The temperature derivative is 0 for vdW, $-\frac{1}{2}$ for RK, and $-\kappa\sqrt{T_r}\,/\,[1+\kappa(1-\sqrt{T_r})]$ for SRK and PR. For PR, with $A=q\beta$ and $B=\beta$, the first of these becomes

$$\ln\varphi = Z-1-\ln(Z-B)-\frac{A}{2\sqrt{2}\,B}\ln\frac{Z+(1+\sqrt{2})B}{Z+(1-\sqrt{2})B}.$$

Because $G^R = H^R - TS^R$ holds by construction, it also serves as a consistency check on the code.

2.4 Total enthalpy and entropy

Absolute values are reported relative to the ideal gas at $T_0 = 298.15$ K and $P_0 = 1$ bar, using an ideal-gas path plus residuals (SVNA Eqs. 6.72, 6.73), with $C_P^{ig}/R = \mathrm{A}+\mathrm{B}T+\mathrm{C}T^2+\mathrm{D}T^{-2}$ (constants of Table C.1, upright to distinguish them from A, B above):

$$H = R\int_{T_0}^{T}\frac{C_P^{ig}}{R}\,dT + H^R, \qquad S = R\int_{T_0}^{T}\frac{C_P^{ig}}{R}\,\frac{dT}{T} - R\ln\frac{P}{P_0} + S^R. \quad (4)$$

3. Program structure and algorithm

The calculator is one self-contained notebook script built from Python dataclasses, NumPy, pandas, Matplotlib and `ipywidgets`. Units are K, bar, $\mathrm{cm^3\,mol^{-1}}$, $\mathrm{J\,mol^{-1}}$ and $\mathrm{J\,mol^{-1}\,K^{-1}}$ ($R = 83.1446\ \mathrm{bar\,cm^3\,mol^{-1}\,K^{-1}}$ for volumes and $8.31446\ \mathrm{J\,mol^{-1}\,K^{-1}}$ for energies).

Table 2: Main building blocks of `pvt_interactive_simulator.py`.

Block	Role
`Species`, `SPECIES`	Database of **36 species**: T_c, P_c, ω (Table B.1), C_P^{ig}/R constants and limit $T_{\max}$ (Table C.1).
`EOSModel`, `MODELS`	$\sigma,\epsilon,\Omega,\Psi,Z_c$ and the functions $\alpha(T_r,\omega)$ and $d\ln\alpha/d\ln T_r$ for vdW, RK, SRK, PR.
`_solve_cubic_Z`	Builds Eq. (2) from $(\beta,q,\epsilon,\sigma)$, finds all roots with `numpy.roots` (no starting guess needed), discards complex roots (tolerance 10^{-9}) and roots with $Z\le\beta$.
`CubicEOS.state(T,P)`	Evaluates G^R/RT, H^R/RT, S^R/R at every root with Eq. (3), picks the root of lowest G^R/RT, and returns Z, $V=ZRT/P$, phase, all roots, H^R, S^R, G^R, $\ln\varphi$, φ, f.
`icph`, `icps`, `H_ig`, `S_ig`	Closed-form ideal-gas integrals (SVNA Eq. 4.8 and its entropy analogue) giving H, S of Eq. (4).
`update_pvt_dashboard`	Widget callback: comparison table (ideal, vdW, RK, SRK, PR) at the chosen (T,P) and six plots.

Algorithm for one state. (1) Read fluid, T, P; compute T_r, P_r, β, q. (2) Solve the cubic, keep the physical roots, and select the root of lowest G^R/RT. (3) Compute V, phase, H^R, S^R, $\ln\varphi$, φ, f, then add the ideal-gas terms of Eq. (4) for H and S. (4) Repeat for every EOS and display the comparison table and plots.

4. Results and validation

The program is checked against (a) worked examples and tables in the book, (b) independent numerical integration of the defining integrals, and (c) the normal boiling points of the book's Table B.1.

4.1 Textbook benchmarks

SVNA Table 13.7 (n-butane, 500 K, 50 bar) is reproduced for all four equations (Table 3); the small differences in the last digit come from the book rounding R to 8.314 and its intermediate values. **SVNA Example 3.9** (RK, n-butane, 350 K, 9.4573 bar) gives $Z_v = 0.8305$, $V_v = 2555.5\ \mathrm{cm^3\,mol^{-1}}$ and $V_l = 133.3\ \mathrm{cm^3\,mol^{-1}}$ (book: 2555 and 133.3).

Table 3: n-Butane at 500 K and 50 bar: SVNA Table 13.7 versus the program (H^R in $\mathrm{J\,mol^{-1}}$, S^R in $\mathrm{J\,mol^{-1}\,K^{-1}}$).

EOS	Z (book)	Z (code)	H^R (book)	H^R (code)	S^R (book)	S^R (code)
vdW	0.6608	0.6610	−3937	−3936	−5.424	−5.421
RK	0.6850	0.6852	−4505	−4503	−6.546	−6.542
SRK	0.7222	0.7224	−4824	−4822	−7.413	−7.408
PR	0.6907	0.6909	−4988	−4985	−7.426	−7.421
Handbook	0.7060	–	−4760	–	−7.170	–

Relative to the Handbook values in the same table, the program's errors are: vdW -6.4% in Z, -17.3% in H^R, -24.4% in S^R; RK -2.9, -5.4, -8.8%; SRK $+2.3$, $+1.3$, $+3.3\%$; PR -2.1, $+4.7$, $+3.5\%$. SRK and PR are clearly the better equations for residual properties.

4.2 Consistency checks

The closed forms of Eq. (3) were compared with direct numerical integration of the defining integrals

$$\ln\varphi = \int_0^P (Z-1)\,\frac{dP}{P}, \qquad H^R = -RT^2\int_0^P \left(\frac{\partial Z}{\partial T}\right)_P \frac{dP}{P}, \qquad S^R = -R\left[T\int_0^P \left(\frac{\partial Z}{\partial T}\right)_P \frac{dP}{P} + \int_0^P (Z-1)\,\frac{dP}{P}\right]$$

(trapezoidal rule, 8000 intervals, central difference with $\Delta T = 10^{-3}T$). All states have $T > T_c$, so a single root exists along the whole path, and agreement is within about 10^{-5} relative (Table 4). In addition, $\max|G^R - (H^R - TS^R)| = 1.3\times10^{-11}$ J mol^{-1} over 2304 states (36 species, 4 EOS, 4 temperatures, 4 pressures), and at $P = 0.01$ bar the program returns $Z = 0.99995$ and $\varphi = 0.99995$ for CO_2 (PR, 300 K), as the ideal-gas limit requires.

Table 4: Closed forms (program) versus numerical integration.

		H^R (J mol^{-1})		S^R (J mol^{-1} K^{-1})		$\ln\varphi$	
Case	EOS	closed	integral	closed	integral	closed	integral
n-C_4H_{10}, 500 K, 50 bar	PR	-4985.00	-4985.04	-7.4214	-7.4214	-0.30653	-0.30653
n-C_4H_{10}, 500 K, 50 bar	SRK	-4821.56	-4821.59	-7.4082	-7.4083	-0.26879	-0.26879
n-C_4H_{10}, 500 K, 50 bar	RK	-4503.03	-4503.07	-6.5424	-6.5425	-0.29631	-0.29631
n-C_4H_{10}, 500 K, 50 bar	vdW	-3935.51	-3935.54	-5.4208	-5.4208	-0.29470	-0.29470
CO_2, 400 K, 100 bar	PR	-2685.04	-2685.05	-5.0484	-5.0484	-0.20016	-0.20016
CH_4, 300 K, 100 bar	SRK	-1640.52	-1640.53	-4.2156	-4.2156	-0.15068	-0.15068

4.3 Vapour pressure at the normal boiling point

Equating the fugacity coefficients of the liquid and vapour roots, $\varphi_L = \varphi_V$, by successive substitution $P \leftarrow P\exp(\ln\varphi_L - \ln\varphi_V)$ gives the vapour pressure P^{sat} from the same $\ln\varphi$ expression as the program. At each normal boiling point T_n of SVNA Table B.1, the exact answer is 1.01325 bar (Table 5). For n-butane at 350 K the book's value is 9.4573 bar: PR gives 9.469 bar ($+0.13\%$), SRK 9.588 bar ($+1.4\%$) and RK 11.414 bar ($+20.7\%$). (This is a check script; the dashboard itself does not compute P^{sat}.)

Table 5: Predicted vapour pressure (bar) at T_n; target 1.01325 bar; deviation in brackets (%).

Fluid	T_n (K)	RK	SRK	PR
Methane	111.4	0.711 (-29.8)	0.959 (-5.4)	0.994 (-1.9)
Ethane	184.6	1.081 ($+6.7$)	0.997 (-1.6)	1.019 ($+0.6$)
Propane	231.1	1.317 ($+30.0$)	1.008 (-0.6)	1.022 ($+0.8$)
n-Butane	272.7	1.518 ($+49.8$)	1.008 (-0.6)	1.017 ($+0.3$)
Nitrogen	77.3	0.862 (-14.9)	0.992 (-2.1)	1.017 ($+0.3$)
Benzene	353.2	1.600 ($+57.9$)	1.007 (-0.6)	1.020 ($+0.6$)
Ammonia	239.7	1.961 ($+93.6$)	0.969 (-4.3)	0.994 (-1.9)
Water	373.15	2.940 ($+190.1$)	0.924 (-8.8)	0.960 (-5.2)

4.4 Dashboard example

Figure 1 shows the six panels produced for the default widget state, CO_2 at 310 K ($T_r = 1.019$) and 80 bar, and Table 6 lists the corresponding table rows. The P–V panel uses PR; the others compare ideal gas, vdW, RK, SRK and PR.

Table 6: CO_2 at 310 K and 80 bar (default dashboard state); all roots are single because $T > T_c$.

Model	Z	V (cm^3 mol^{-1})	$\ln\varphi$	φ	f (bar)	H^R (kJ mol^{-1})
Ideal gas	1	322.2	0	1	80.00	0
vdW	0.3764	121.3	-0.3804	0.6836	54.69	-4.621
RK	0.4092	131.8	-0.4160	0.6597	52.77	-5.288
SRK	0.4384	141.2	-0.4102	0.6635	53.08	-5.755
PR	0.4117	132.6	-0.4470	0.6396	51.17	-5.814

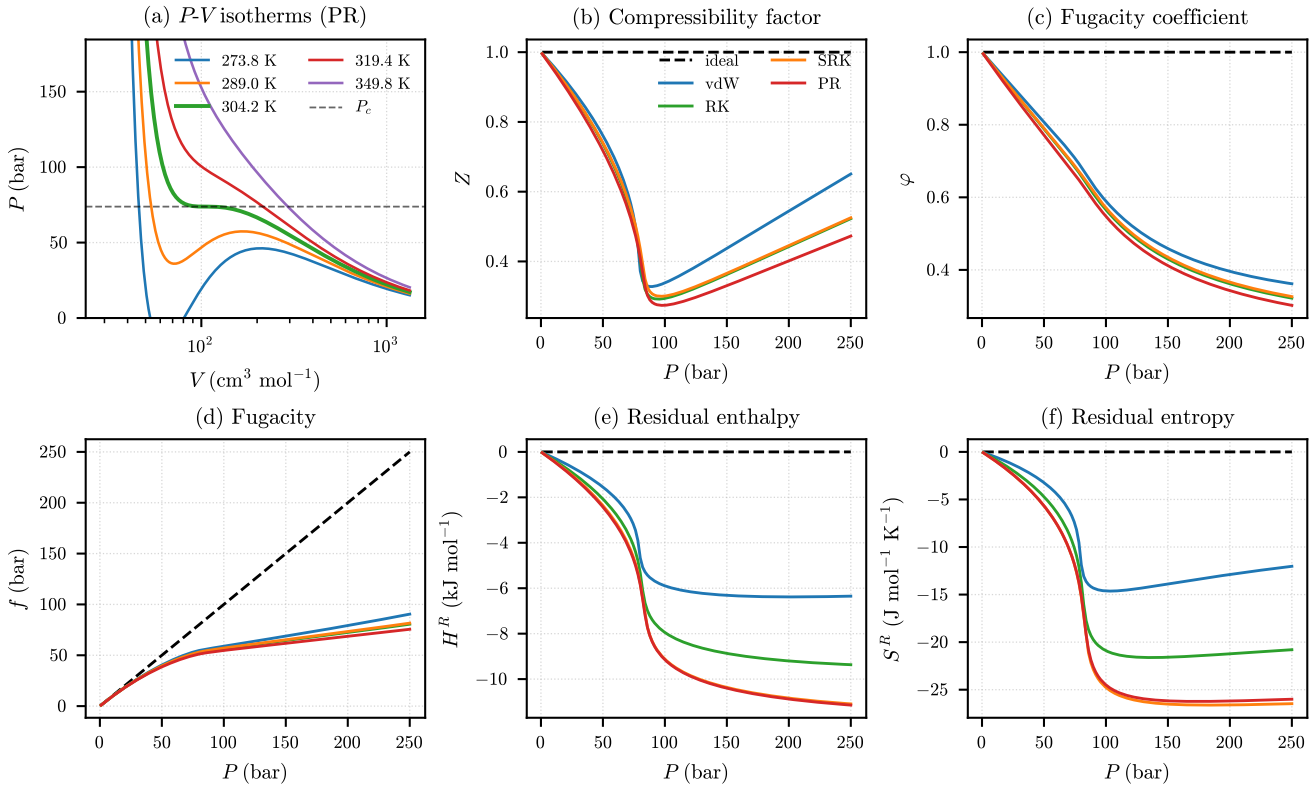


Figure 1: Dashboard output for CO_2 at 310 K: (a) PR P–V isotherms at 0.90–1.15 T_c with the critical isobar; (b)–(f) Z, φ, f, H^R and S^R against pressure for the ideal gas and the four cubic equations.

5. Discussion and conclusions

Accuracy. SRK and PR give the best residual properties (Table 3), and PR gives the best vapour pressures, within 1% for non-polar fluids from nitrogen to benzene. RK is acceptable for gas-phase Z but has no acentric factor, so its vapour pressures are poor, as Table 5 and the n-butane result at 350 K show. vdW is only qualitative.

Limitations. (1) Near the critical point, as for CO_2 at $T_r = 1.019$ in Figure 1, Z falls steeply with pressure and all cubics are least reliable. (2) Polar and associating fluids are represented less well: PR misses the vapour pressure of water by -5.2% and ammonia by -1.9% because the standard $\alpha(T_r)$ was developed mainly for non-polar fluids. (3) Z_c is fixed by the equation (1/3 for RK and SRK, 0.307 for PR), whereas the true value for n-butane is 0.274 (Table B.1), so liquid volumes are only approximate. (4) The C_P^{ig} fits are valid from 298 K to $T_{\max}$; absolute H and S below 298 K, or above $T_{\max}$, are extrapolations, and the sliders do not restrict T to this range. (5) The label *Supercritical* is given to every state with $T \geq T_c$, including low-pressure gas. (6) Only pure components are treated.

Conclusion. The Real Gas Calculator meets all three stages: it solves the generic cubic for all physical roots, selects the stable one from G^R/RT, and returns Z, V, H^R, S^R, H, S, φ and f for 36 gases in an interactive notebook. It reproduces SVNA Table 13.7 and Example 3.9, its analytic residual properties match numerical integration, and PR and SRK are recommended for engineering work on non-polar gases. Run it in a Google Colab notebook for the intended interactive experience.

References

[1] J. M. Smith, H. C. Van Ness, M. M. Abbott and M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill, 2018: Table 3.1, Sec. 3.6, Example 3.9, Eqs. 6.72–6.73, Sec. 13.6, Example 13.5, Table 13.7, Sec. 13.7, Tables B.1 and C.1.
[2] O. Redlich and J. N. S. Kwong, *Chem. Rev.* **44** (1949) 233–244.
[3] G. Soave, *Chem. Eng. Sci.* **27** (1972) 1197–1203.
[4] D.-Y. Peng and D. B. Robinson, *Ind. Eng. Chem. Fundam.* **15** (1976) 59–64.